

Understanding and Avoiding Plagiarism: An Author's Guide

IEEE Computer Society Course
Material

Digital Text: Easy Copying, Easy Plagiarism, Easy Detection

- With So much digital contents available in the Internet and highly searchable nature of digital text
 - Easy to detect uncredited copying original documents/materials
 - No of reported incidents has increased steadily since 2005
- IEEE defines plagiarism
 - Reuse of someone's prior processes, results, or words without explicitly acknowledging the original author or source.

Plagiarism in any form, at any level, is unacceptable and is considered a serious breach of professional conduct, with potentially severe ethical and legal consequences.

Digital Text: Easy Copying, Easy Plagiarism, Easy Detection

- Is there any established measurement that can be used to determine when plagiarism has taken place? In fact plagiarism is plagiarism, but there are degrees of plagiarism:
 - Inappropriate reuse an entire article, or a section of an article, or a page, a paragraph or just a sentence.
 - Even paraphrasing done improperly can be characterized as plagiarism

Using the Archive: No Exceptions to the Rule

- The fundamental nature of scientific/technical writing is that it is closely based on the archival literature.
- New work typically refers back to any seminal work that has already been published in order to establish a necessary level of authentication and validation.
- If new work depends so much on the close use of the archive, is it reasonable to ask if exceptions should be made for scientific or technical writing where the rules against plagiarism are concerned?
- Some authors believe that technical writings are not literary writing, it should be acceptable to use a “certain amount” of someone else’s text without having to indicate the specific text, especially when a citation appears on the same page or in the reference section.

Using the Archive: No Exceptions to the Rule

- The same kind of thinking might also assert that the use of quotation marks or indented text to identify someone else's text would interrupt the flow of the writing and interface with the reader's comprehension of the work.
- Shouldn't the nature of technical/scientific writing allow exceptions to the formal use of quotation marks and/or indented text

There are no valid exceptions to plagiarism.

How to Avoid Committing Plagiarism

- If author 'A' would like to use text, charts, photographs, or other graphics from author B's original material, then author A must do two important things:
 1. Clearly indicate the reused material by using quotation marks or indented text and provide a full reference to the original material (publication title, author name, article title, etc.)
 2. Obtain written permission from the publisher or, if the reused material has not been published, obtain written permission from the original author.

How to Avoid Committing Plagiarism

- Amount or quantity does NOT play a role in defining plagiarism
- Remember: Plagiarism is plagiarism
- However, the amount of plagiarized materials DOES play an important part in determining the most appropriate penalties when plagiarism is discovered and reported.

How to Avoid Committing Plagiarism

- In general, the appropriate use of credit notices or references are not sufficient to avoid a charge of plagiarism if quotation marks or offset text within the body of an article have not been used to identify the specific material being copied.
- Similarly, improper paraphrasing can leave an author open to a charge of plagiarism if he or she has changed only a few words or phrases or has only rearranged the original sentence order.
- Again, even proper paraphrasing of the original text can lead to a charge of plagiarism if the original source is not properly cited.

A Quick Review

- It is important for all IEEE authors to recognize that plagiarism in any form, at any level, is unacceptable and is considered a serious breach of professional and ethical conduct.
- If an author neglects to cite the copied work properly, either by choice or by mistake, the author is committing plagiarism.
- When authors reuse someone else's work and make it appear to be their own, they deny the original author credit for her or his contributions to the research community and to society.
- Plagiarism is a type of *copyright infringement* and as such may also expose an offender to legal liability.

IEEE Approach to Combatting Plagiarism

A Two-Pronged Approach

- In an effort to reduce the amount of plagiarism in IEEE's submitted and published contents, as well as plagiarism of IEEE-copyrighted material in non-IEEE journals, the IEEE IPR office has taken a two-pronged approach to manage this challenge.
- The first is to **detect** plagiarism at the earliest possible opportunity, which is when an article is first submitted to IEEE.
- The second approach is to **resolve** plagiarism matters as they are discovered after a problem article has already been published in IEEE Xplore.
- Also, as part of plagiarism resolution, IEEE will contact other publishers regarding possible copyright infringement when IEEE material is copied and re-published without credit in non-IEEE journals.

The First Approach: Plagiarism Detection

- Plagiarism detection is done primarily through the use of CrossRef's Similarity Check Service (CrossCheck) by IEEE volunteers (Editors/Conference Organizers) when articles are first submitted.
- In addition to this computer-assisted form of plagiarism detection at submission, many instances of plagiarism are also found by reviewers and editors in the early stage of peer review.
- IEEE Xplore users (readers and authors) are also helpful in reporting problems when they encounter plagiarism in an already published work in IEEE Xplore.

The Second Approach: Plagiarism Resolution

- Plagiarism found in IEEE Xplore requires a formal review by the editor of the publication or the organizer of the conference in which the alleged plagiarism was found and, in case involving very serious plagiarism, by the Vice President of the IEEE Publication Services and Product Board (PSPB) and the Chair of the PSPB Publishing Conduct Committee.
- If, at the end of the formal plagiarism-resolution process, it is determined that the author has in fact inappropriately reused another author's work, the offending author's name and email address are added to the Prohibited Author Listed (PAL) which is available to all editors for use at the time all articles are first submitted.

Consequences

- If an editor should subsequently identify a banned author while reviewing a newly submitted article, the author's article will then be rejected immediately.
- Authors will not be permitted to resubmit the rejected article until the banned period has ended.

A Quick Review

- IEEE uses a two-pronged approach to combatting plagiarism; to detect plagiarism at the earliest possible opportunity, and to resolve plagiarism matters as they are discovered.
- Plagiarism detection is done at manuscript submission through the use of CrossRef's Similarity Check Service by IEEE publication volunteers.
- Plagiarism found after publication in IEEE Xplore requires a formal review by the publication editor or the conference organizer. Serious plagiarism will require a review by the Vice President of the IEEE PSPB and the Chair of the PSPB Publishing Conduct Committee.
- Serious level of plagiarism may result in an author being added to PAL . When an editor identifies a prohibited author, the author's article will then be rejected accordingly.

Similarity Check, a Premier Plagiarism Detection Tool

- To help ensure the integrity of IEEE's high-quality publications, the IEEE Board of Directors has required that all contents submitted to IEEE publications must be screened for plagiarism.
- To assist in fulfilling this policy, IEEE provides its editors and conference organizers with direct access to Similarity Check, an easy to use and powerful plagiarism-detection tool that compares newly submitted articles against a very large database of published technical articles (in addition to billions of web pages) and returns a summary report that highlights any detected similarity to previously published work.
- Editor can then follow up to review and possibly suspend articles with high similarity scores as needed.

Effective Plagiarism Detection

- Similarity Check's significant advantage over other services is that it includes the indexed full-text content of participating Similarity Check member publishers.
- It not only identifies similarity, but it also provides a similarity score that helps to establish how much of the work is original and how much is copied from another source.
- The IPR office staff will provide feedback to editor and conference organizers whenever plagiarism is identified during the review process so that the publishing volunteers can more confidently decide to pursue a formal plagiarism-resolution case or to advise the author about other corrective actions.

Effective Plagiarism Detection

- The goal of early plagiarism detection is twofold:
 - 1.To identify at the earliest possible time an article that contains plagiarized content, thereby ensuring that corrective actions are taken before the article is published.
 - 2.To reduce the number of formal plagiarism cases that occur after publication in IEEE Xplore.

Effective Plagiarism Detection

- Articles are uploaded to Similarity Check upon submission by a number of possible methods.
- Most submissions to IEEE periodicals are submitted to Similarity Check through ScholarOne Manuscripts (S1M) submission platform during peer review.
- Articles submitted to conferences or periodicals that do not use S1M may be uploaded to Similarity Check by another article submission vendor contracted by the conference or periodical, or by the conference organizer using IEEE Similarity Check portal.

A Quick Review

- IEEE requires that all contents submitted to IEEE publications must be screened for plagiarism.
- IEEE provides its publications volunteers with direct access to Similarity Check, a plagiarism-detection tool that compares submitted article against a very large database of published technical articles or web pages.
- The IPR office has a staff team dedicated to review the results of all submissions to Similarity Check, and to provide feedback to editors and conference organizers whenever plagiarism is identified during the review process.
- The goal of screening for plagiarism at submission is to identify at the earliest possible time an article that contains plagiarized content, and to reduce the number of formal plagiarism cases that occur after publication in IEEE Xplore.
- Similarity Check is used by article submission vendors such as ScholarOne, and is available as a standard service in the IEEE Similarity Check Portal.

The Impact of Plagiarism

Impact on Offending Authors

- Plagiarism is a serious form of publishing misconduct that can have far-ranging and severe consequences for the offending author.
- Anyone who has been found guilty of plagiarizing an original author's work will face one or more of the following.
 - Expulsion from university degree programs
 - Job loss
 - Revocation of advanced degree
 - Damaged reputation
 - Prohibition from future publication with IEEE

Impact on Original Authors and Readers

- The original authors may take personal offense to the blatant and disrespectful theft of their work, which sometimes represents several years of dedicated effort.
- However, beyond its personally offensive nature, plagiarism also causes logistical problems for many readers, researchers and other users.
- When an undetected plagiarizing article is published and enters the archive, readers and especially researchers who discover the article may mistakenly cite the plagiarizing article as the original source of valuable information. This in turn, can cause a long-lasting *domino effect* of reference inaccuracy and bibliography error.

Impact on Coauthors

- When plagiarism is found in an IEEE article, many IEEE societies agree that all authors of the article should be held responsible for the misconduct.
- An author who may be thinking about inappropriate copying of published material should keep in mind that such actions could affect the future career and reputations of coauthors as well.
- It is equally important that the coauthor of an article check for plagiarism before agreeing or submitting.
- If an author's name is on the article, he or she will most likely be held accountable for any plagiarism that is discovered.

Referencing and Citation Guidelines

References & Permissions

- In scholarly writing, it is almost always necessary for authors' research to include other authors' research results, or to reuse portions of their own previously published articles, in order to support the new article being prepared for submission.
- However, it is essential for authors to:
 - Give full credit to any reused, previously published work.
 - Identify clearly in the new article those portions that have been reused from an original authors' article.

Authors must also obtain formal permission from the publishers of the source materials, or obtain written permission from the author if the reused material has not yet been published

Quotation Marks, Citations & Cover Letters

- Similarly, if authors wish to reuse content from their own previously published articles, they must clearly identify all reused contents by using quotes or indents and provide full citations to the original publication of the article.
- If any previously published or submitted material is used as the basis for a new submission, authors must also state this in a cover letter to the editor and in the text of the article and explain how the new submission differs from the previously published work

Building References

- Sources of published work that have been used to develop the article should be listed separately at the end of the article in the reference section.
- Generally, the sources must be numbered, and each source in the reference list must be given a separate numbered entry, and should be arranged in the order they have been used in the article.
- If content from a source is used more than once within the article, then separate notations should be made to the same referenced source.

Important Guidelines

- When text from another author's article is reused, the text must be set off by either quotation marks or be clearly indented from the surrounding text and include an in-text-reference number that corresponds to the same number in the reference section of the article where citations to the original source can be found.
- Similarly, when paraphrasing or summarizing material from an original source (including a discussion of ideas, processes, arguments, or conclusions), authors must add in-text numbers that corresponds to the same number in the reference section of the article where credit to the original can be found.
- When reproducing figures, graphics, tables or data, authors must refer to the original source when reusing or revising

A Quick Review

- Authors must give full credit to any reused, previously published work and identify clearly in the new article any portions that have been reused from an original author's article.
- They must also obtain formal permission from the publishers of the source material.
- If authors reuse content from their own previously published article, they must clearly indicate all reused content by using quotes or indents and provide full citations to the original publication of the article.
- If any submission published or submitted material is used as the basis for a new submission, authors must also indicate this in a cover letter to the editor and in the text of the article and explain how the new submission differs from the previously published work.
- Sources that have been used to develop the article should be listed separately at the end in a reference section. When references are noted in the text, they must appear next to the reused content.

Publishing Ethics in Scholarly Publishing

Principles of Scholarly Publishing

- Scholarly publishing is founded on several principles that are cited to the publishing process and to the overall value of the published material.
- IEEE has developed and issued a set of core principles called the “principles of Scholarly Publishing” that are important to consider in any discussion about the business of publishing and the problems associated with plagiarism and other forms of misconduct.
- For example, one of the core principle provides an all-encompassing commitment to the public benefit:
 - “Not-for-profit learned societies have the unique and overarching goal of conducting their activities, including publishing, not for the benefit of any individual or group of individuals, but rather to provide services that benefit the global public in both direct and indirect ways.”

Principles of Scholarly Publishing

- Similarly, another core principle speaks to the essential importance of preserving the archival digital content:
 - “Society also benefits from the commitment and experience that scholarly publishers bring to the challenge of archival preservation of electronic content to ensure the availability of published literature for future generations.”
 - These are not causal statements. They are at the core of IEEE’s publishing efforts and must be understood as uncompromising where the public’s interests in open and honest communication are concerned.

The Value of the Archive

- As noted, several of the principles invoke the value of the published archive, or the version of record.
- An accurate and properly indexed archive is extremely important for scholarly publishing because authors must depend on work that has been done in the past in order to move forward in their own careers.
- Since the value of an accurate archive is high, it is important to protect and maintain the value by insisting authors follow the best scholarly practices involving the proper reuse of another author's work through the use of citations and references.
- The principles help shape the philosophy and set of ethics that guide how we treat our authors and members.

The Value of the Archive

- IEEE's role not only as a scholarly publisher but also as keeper of a substantial engineering and scientific archive requires that ***we unmistakably assert to all submitting authors that we will do our utmost to keep the archive accurate at all times.***
- IEEE's commitment to "Advancing Technology for the Benefit of Humanity" can be applied here, in that the archive is not only essential to individual authors within a given field, but also has value that can be extended to the global public, directly and indirectly.
- **These core values make it plain that IEEE will remain steadfast in discovering and removing plagiarism from IEEE's entire publishing programs**